STUDENT NAME:	Francisco Sandoval Hernández
SUPERVISOR NAME:	Moisés Martínez Muñoz

TUTORING SESSION Nº:

☐ 1. Start
☐ 2. State of the Art / Previous Research
☐ 3. Project Plan
☒ 4. Project Development
☐ 5. Technical Work Closure
☐ 6. Report Review
☐ 7. Other: Specify

DATE: 20/02/2026
START: 19:00
END: 20:00

DISCUSSED TOPICS:

During this tutoring session, the development progress of the landmark-based recognition pipeline was reviewed. The student presented the WLASL30 benchmark and the first sequence-modelling experiments based on MediaPipe hand landmarks. Several candidate architectures were considered, including LSTM, GRU, GRU with enriched features, GRU with temporal attention, and a Transformer baseline.

The supervisor and student discussed the results of the initial architecture comparison. The GRU+Attention model showed the best balance between performance, model complexity, and suitability for the available data regime. The attention mechanism was considered useful because sign language clips contain preparation, execution, and recovery frames, and not all frames contribute equally to recognizing the sign.

The limitations of the evaluation protocol were also discussed. Since the WLASL30 test set was small, it was agreed that reporting only a single result would not be sufficiently robust. The need for repeated experiments, additional validation, and confusion-matrix analysis was established to better understand whether the model was learning stable patterns or only fitting a specific split.

The use of Test-Time Augmentation was also considered. Although it could improve some numerical results, it was agreed that it should not be included in the final evaluation unless it clearly improved generalization. The priority would be to keep the evaluation protocol simple, reproducible, and easy to interpret.

Finally, the next development stage was planned. The selected model would be tested with different sequence lengths and later applied to larger WLASL vocabularies to study scalability. The supervisor also suggested analyzing the gap between small and large vocabularies, since the main difficulty seemed to be the combination of many classes with few examples per class.

ACUERDOS PARA LA SIGUIENTE REUNIÓN:

Item	Agreement / Task	Person in Charge	Deadline
1	Complete the architecture comparison on WLASL30.	F.S.H	06/03/2026
2	Run the sequence-length sweep with 32, 64 and 96 frames.	F.S.H	06/03/2026

3	Prepare confusion matrices and per-class analysis for the selected model.	F.S.H	06/03/2026
4	Start the vocabulary-scaling experiments beyond 30 classes.	F.S.H	06/03/2026

SCHEDULED DATE OF THE NEXT MEETING: **10/04/2026 – 19:00**

Supervisor's Comments:

Supervisor's Signature:	Student's Signature:
 Moisés Martínez Muñoz	